Figure Legends

Figure 1. Stromal cell-state annotation and metadata coverage

(A,B) Original fibroblast and endothelial-lineage UMAPs colored by Celltype_general. Each point is one of 160,000 stratified representative cells per lineage; full source counts are 5,916,760 and 2,239,277 cells, associated with 4,818 and 3,553 sample labels. Broad colors follow the displayed source legends. The endothelial-lineage collection retains the source MCs category. (C,D) The same plotted cells colored by Tissue_from_Celltype, the terminal underscore-delimited Celltype annotation suffix. These 12/11 suffix categories, including pan, are not equivalent to original anatomical tissue. (E) Full-cell and distinct-sample counts within suffix categories; fibroblast bars extend left and endothelial bars right, both encoding positive counts. Samples can occur in several categories. (F) Counts by recorded Age_group, including Unknown. Unverified age-bin boundaries were not invented. (G,H) Broad-state marker dot plots: dot size is detection fraction and blue intensity is gene-wise min–max-scaled mean stored expression. Unavailable fibroblast markers PI16, LUM and IL33 are omitted. (I,J) Signature means by broad annotation: colors are column z-scores and printed values are unscaled state means. Annotation colors and continuous score colors have different meanings. All counts/expression summaries use full-source cells; no inferential significance is attached. Marker/signature support shares the annotated data and is descriptive. Source: F1 fig1_20261103_v2.

Figure 2. Supplied clock overview and source-fold-4 performance

(A) Complete supplied workflow from TabulaStromal_Workflow_V6.pdf. This is conceptual artwork: illustrative numbers, donor icons, the healthy-only label, biological-age wording and GWAS/TF sketches do not establish corresponding empirical findings in the audited study. Actual targets are recorded chronological ages, and a verified donor crosswalk is unavailable. (B–D) R^2, MAE and label MAD for 13 models in Fibroblast and Endothelial Train/Valid/External sets. Each model's K was selected internally using source fold 4. Red/blue R^2 colors are centered on zero; darker blue MAE/MAD denotes larger values, not necessarily better performance. MAD=mean absolute deviation of Age from its mean; it is not prediction error. ElasticNet is highlighted for the later interpretable-clock branch, without implying global superiority. (E,F) ElasticNet K=20,000, source fold 4, with each point representing a unique sample. Train/Valid sample counts are 513/127 for Fibroblast and 351/86 for Endothelial. Gray circles, orange triangles and green squares denote Train, Valid and External; dashed lines indicate identity. The external sets contain 93 and 77 samples and use FT30 → ALL100, including adaptation samples. R^2/MAE are 0.410/8.95 years and 0.218/8.77 years, respectively. These are mixed apparent/unseen results, not untouched external tests. Legends were recomputed from displayed pairs. No error bars or model-superiority significance tests are shown. The source y-axis wording predicted biological age must be interpreted as predicted chronological age and revised before publication. These models differ from the full-development final clocks in Figure 3.

Figure 3. Final EN prediction protocols and exploratory stability

Panels are taken from FINAL_v2; Fibro precedes Endo within each paired analysis. (A,B) Original-feature versus stable-feature EN outer-validation MAE. Each dot is one fold, gray lines join matched held-out evaluations, and bars show arithmetic means without inferential error bars. K is fixed at 20,000/10,000 after the earlier EN benchmark; the five-fold estimates are conditional on that selection. Development sample/group counts are 640/634 and 437/431. (C,D) Histograms of full-development Celltype_Gene selection frequency across 300 group bootstraps, using a logarithmic count axis. Dashed/dotted thresholds are 0.70/0.75. The primary stable sets contain 438/23 pairs and 416/23 unique genes across 5/1 detailed cell types. (E,F) Pooled original-EN OOF predictions from fold-trained models, and predictions of the final full-development model on merged external samples without adaptation. Each point is a sample; internal circles and external triangles are distinguished in the displayed legends. Pooled OOF metrics need not equal the arithmetic mean of fold metrics. External n=93/77. No external labels were used for the primary fit. (G,H) Stable features ranked by absolute mean standardized bootstrap EN coefficient, displaying at most 50 entries; Fibro shows 50 and Endo all 23. Blue/gray bars indicate positive/negative mean coefficient, not causal pro-/anti-aging status. (I,J) Candidate-normalized stable-pair fractions by detailed Celltype, retaining zero-stable candidates; denominators are variance-eligible candidate pairs. (K,L) Genes ranked by maximum constituent-pair selection frequency, displaying at most 50 genes. This is not a gene-level bootstrap selection probability. (M,N) Four distinct evaluation scopes: noadapt ALL100, noadapt heldout70, adapted heldout70, and adapted ALL100. Labels report n, R^2 and MAE; no error bars are implied. FT30 uses 28/23 fitted samples and 65/54 held-out samples. ALL100 includes fitted observations and is mixed apparent/unseen. (O,P) Up to 12 global functional terms with observed overlap, ordered by FDR. The x-axis is −log10(BH FDR), point area depends on overlap count, and the dashed line marks FDR=0.05. Full testing includes zero-overlap size-eligible terms within library/scope. (Q,R) Counts overlapping the eight supplied aging gene lists with printed FDR. All lists remain provided_unverified; statistical overlap is not authentication of official signatures. Stable-set interpretation is exploratory and distinct from the unreduced final predictor.

Figure 4. Historical CellAge integration of cellular states and age-related analyses

Fibroblast panels are on the left and endothelial-lineage panels on the right. (A,B) Existing Atlas UMAPs for visualization, stratified by GSE, source-designated donor/sample and subtype; broad annotation defines color. Display sampling does not change full score-analysis denominators. (C,D) Mean sample-designated donor hUSI by detailed Celltype, with equal source-unit weighting and 95% intervals from 1,000 bootstrap resamples. Primary score aggregation requires at least 20 scored cells per sample/tissue/subtype. Native scores are dataset-min–max normalized and are neither ages in years nor calibrated cross-lineage quantities. (E,F) Adjusted within-subtype age slopes with source-unit-cluster 95% confidence intervals. Models account for available tissue, sex, GSE, disease, treatment, platform and QC proxies; warm-colored points identify BH $q<0.05$ where present. Fifteen/six primary adjusted models were identifiable, with no $q<0.05$ findings. Full n and non-identifiability reasons remain in the source tables. (G,H) Six standardized SenCID identity-recommendation dimensions; these are not senescence intensity or probability. (I,J) Exploratory comparison of hUSI age slopes with signed coefficients of the older EN model, substituted because adjusted clock associations were unavailable or not identifiable. These are not final-clock validation panels. (K,L) Exploratory pathway overlap of the older stable feature sets using actual subtype-tested candidate

backgrounds and the historical CellAge correction family; at most 14 representative terms are shown. RecSID does not validate genuine SenCID intensity, and enrichment is not causal evidence. The original CellAge donor terminology reflects its sample=donor assumption, which remains unresolved against the current cross-study identity audit. Source run: 20261030_220917_407544.

Extended Data Figure 1. Detailed annotation and pooled composition

(A,B) Detailed Celltype UMAPs within broad fibroblast/endothelial-lineage groups, comprising 32/38 labels. Each broad-state subplot shows only its own cells, drawn from the same 160,000 representative cells per lineage as Figure 1. Labels retain original names and leader lines point to displayed cells. Separate subplot coordinate ranges differ, precluding quantitative distance comparison between subplots. (C,D) Pooled broad-state cell proportions within Celltype-label suffix categories. (E,F) Corresponding proportions within stored age groups, including Unknown. Proportions sum to one within category and are cell-weighted, not equal-donor estimates. Broad colors follow Figure 1. There are no inferential error bars. Source: extended_figure1_subtype.

Extended Data Figure 2. Figure-marker signature agreement and discordant clusters

This legend refers to the 6 November figure-marker version, not the older baseline mean/module analysis. (A) Fibroblast UMAPs colored by ECM, AP and four broad-subtype signature means. (B) Endothelial-lineage UMAPs colored by AP, Tip, Stalk and six broad-subtype signatures. Scores use the recorded global marker lists without per-cluster adjustment; color scales within the source panels should not be interpreted as calibrated across different signatures. (C,D) All 55 fibroblast and 44 endothelial annotation×Leiden records, each titled with the original Celltype_CLeiden identifier. Bars show unweighted mean stored log expression across the corresponding cluster cells for each signature. Broad annotation sets background color; bar color denotes the scored signature. Black outlines and enlarged numbers identify the observed maximum. Daggers mark disagreement with the reference annotation, retained for 4/12 records. Cluster agreement is 51/55 and 32/44. Source records give full-cell n, detailed reference composition and exact values; plotted UMAP sampling does not change bar denominators. No p-value or error bar is implied by the maximum. Marker selection was supervised against marker-informed labels, and retained GSEs were not newly blinded. GSE-equal retained concordance and bootstrap intervals are reported in Results 1. Source basename ExtFig1_SignatureScore is mapped to Extended Data Figure 2 here.

Supplementary Figures 1 and 2. Stored fibroblast filtering comparisons

Supplementary Figure 1 contains 24 TISCH datasets in four blocks (A–D, six datasets each); Supplementary Figure 2 contains 32 HUSCH datasets in four blocks (A–D, eight datasets each). Within each dataset column, the eight rows show recall, accuracy, precision, F1, reference-label UMAP, and saved Marker, DEG(only) and DEG(like) UMAPs. Curves use the stored evaluation table across thresholds 0.1–0.9; saved-filter UMAP labels are not assigned an unrecorded operational threshold. Reference labels are Celltype_curated for TISCH and Celltype_level3 for HUSCH. Up to 12,000 fixed-seed stratified cells per dataset are shown with the same IDs across all four UMAP views, while counts and metrics use all source cells. Display legends distinguish reference and filtering colors. No error bars or independent-donor inference are attached. The

saved objects provide AveExp comparisons; unverified logFC/100-DEG/150-DEG analyses were not added. Source basenames Extended_Figure1_TISCH/HUSCH have separate manuscript numbers in the manifest.

Source mapping: four-module figure inventory. Figure 4 uses the large_text display revision of the same audited CellAge run.